DT 111 · TOPIC 8

Contemporary Issues in AI

Generative AI, synthetic media, agents, accessibility, work, governance and responsible use

Mkwawa University College of Education

BSc Data Science and Artificial Intelligence

Learning outcomes

- Explain generative AI and large language models at a beginner level.
- Recognise hallucinations, deepfakes and information-integrity risks.
- Describe AI agents and the additional risks of tool use and autonomy.
- Discuss AI in education, work, accessibility and public life.
- Apply responsible-use principles to contemporary AI systems.

Contemporary AI: new capabilities and new risks

AI systems can now generate content, work across multiple media and use external tools. These capabilities create both opportunities and new responsibilities.

Generative AI creates new content

Text

Answers, summaries, drafts and dialogue.

Images

Generate or edit visual content.

Audio

Speech synthesis, transcription and music.

Video

Synthetic scenes, avatars and editing.

Code

Generate, explain and modify software.

Large language models: a simple mental model

Tokens

Text is processed as smaller units rather than entire ideas.

Context

The prompt and previous conversation influence the next output.

Probability

The model predicts plausible continuations from learned patterns.

Generation

Repeated prediction produces coherent text, code or other sequences.

Example: language-model generation

1

Prompt

“Explain EDA to a beginner”

2

Context

Course language and earlier text

3

Token prediction

Select likely next units

4

Response

Generate a structured explanation

5

Verification

Check facts and suitability

Fluency and factual reliability

What the model can do well

- Produce fluent explanations
- Reformat and summarise text
- Generate examples and drafts
- Combine information in useful ways

Why verification is still needed

- Can invent unsupported facts
- Can misstate citations
- May reproduce bias
- May sound confident when wrong

Hallucination

Meaning

A generated statement that is unsupported, fabricated or incorrect while appearing plausible.

Why it happens

The model is generating likely content, not consulting a perfect database of truth.

Response

Verify important claims against reliable sources, especially in high-stakes contexts.

Synthetic media and deepfakes

Opportunity

Creative production, localisation, accessibility and simulation.

Risk

Impersonation, fraud, misinformation and reputational harm.

Digital literacy

Consider source, provenance, context and independent verification.

Generative AI in education

Useful support

Explanation, brainstorming, practice questions, feedback and language assistance.

Academic integrity

Learners should follow institutional rules and represent their own work honestly.

Learning risk

Over-reliance can reduce independent thinking and skill development.

Privacy

Sensitive student or research data should not be shared carelessly.



AI agents

An AI agent can pursue a goal through multiple steps, potentially using external tools such as search, files, calendars, code or databases.

A simple agent loop

1

Goal

Define desired outcome

2

Plan

Choose a sequence of actions

3

Use tool

Search, calculate or update

4

Observe

Read result or environment

5

Continue / stop

Adapt until complete

Why tool-using AI needs stronger controls

Capability

- Can perform multi-step tasks
- Can interact with external systems
- Can automate routine workflows

Additional risk

- Wrong actions can have real consequences
- Permissions and access matter
- Security and human confirmation may be needed

AI and the changing workplace

Automate tasks

Some routine activities can be completed with less manual effort.

Augment people

AI can support drafting, analysis, search and decision preparation.

Create new tasks

Prompt design, verification, governance and AI-assisted workflows.

Skills adapt

Domain knowledge, critical thinking and digital judgement become more important.



AI for accessibility and inclusion

AI can remove barriers when systems are designed with users rather than merely for them.

Assistive AI examples

Speech-to-text

Convert spoken language into readable text.

Text-to-speech

Read text aloud through synthesised speech.

Visual assistance

Describe images or recognise objects.

Alternative interaction

Voice, text, touch or other accessible control modes.

Personalised learning

Potential

Adapt pace, difficulty, feedback or format to learner needs.

Risk

Poor data or rigid predictions can label learners unfairly.

Good practice

Keep teachers and learners in control and allow adjustment or override.

Mobility and accessibility

Route planning

Include step-free access, surface conditions and accessible entrances.

Navigation assistance

Provide multimodal directions through text, audio or visual cues.

Co-design

People with lived experience should help define requirements and test the system.

AI for emotional well-being

Appropriate low-risk support

- General information
- Journaling or reflection prompts
- Resource signposting
- Escalation to human support

Unsafe overreach

- Autonomous diagnosis
- Replacing emergency or professional care
- Overconfident medical advice
- No escalation pathway



Governance and regulation

Contemporary AI requires organisational policies, technical standards, human-rights safeguards and legal accountability.

Contemporary governance questions

Transparency

Can users understand when and why AI is used?

Accountability

Who is responsible for harmful or incorrect outcomes?

Intellectual property

How are copyrighted inputs and outputs handled?

Information integrity

How are deepfakes and misleading generated content addressed?

Environmental and cybersecurity concerns

Computational cost

Large models can require substantial energy, hardware and data-centre resources.

Cybersecurity misuse

AI can assist phishing, social engineering or code generation for malicious purposes.

Security defence

AI can also support detection, analysis and response when used responsibly.

Responsible use of contemporary AI

1

Define purpose

What problem is being solved?

2

Choose data carefully

Protect privacy and representation

3

Verify outputs

Check facts and uncertainty

4

Control actions

Use permissions and confirmation

5

Monitor impact

Learn from errors and feedback

Topic 8 key ideas

Generative AI can create useful content, but fluent output is not automatically factual.

Synthetic media increases the importance of verification and information integrity.

AI agents can act through tools, increasing the need for permissions, security and oversight.

AI can support accessibility and learning when users retain control and systems are co-designed.

Contemporary AI affects work, governance, security, the environment and society - responsible use must evolve with capability.